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13 Nitrogen Dynamics in Soil-Plant Systems

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Nitrogen plays a key role in plant nutrition. It is the mineral element required in the greatest quantity by cereal crop plants and it is the nutrient most often deficient. As a result of its critical role and low supply, the management of N resources is an extremely important aspect of crop production (Novoa & Loomis, 1981). Nitrogen is currently the most widely used fertilizer nutrient and the demand for it is likely to grow in the foreseeable future.

Despite this large investment in fertilizer N, the efficiency with which crops use it is poor. Allison (1965) suggested the average recovery of fertilizer N in the aboveground parts of crops is about 50%. Power (1981) has indicated that the general range of recovery in plant parts is between 20 to 90% of the fertilizer applied.

The N that is not recovered by the crop may be lost to the atmosphere through volatilization of ammonia, denitrification, or leaching, or made unavailable to the plant through immobilization in the soil. It may also become inaccessible to the plant through lack of water. The magnitude of each of the various transformations affecting the use of N is influenced by many climatic, edaphic, and agronomic factors. The myriad of transformation pathways and the multitude of factors affecting transformation rates renders N as one of the most complex plant nutrients to study. Quantifying these factors and predicting the response to added N is, thus, a very difficult task. The fraction of N that is lost from the cropping system is a source of some of the environmental pollution associated with fertilization. Thus, for economic and environmental reasons, a major thrust of current fertilizer research seeks to improve the efficiency of its use.

Traditionally, information used to generate a response curve is obtained from field experiments that test several increasing dressings of each nutrient to show how the fertilizer effects crop yields. This response information then forms the basis for deciding an optimum application rate. Regression models of fertilizer response, however, are static in nature and are unable to account for within-season variability in the supply of either water or nutrients.

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Computer simulation models that are able to capture the nuances of weather and the effects of various soil properties and agronomic practices on nutrient dynamics and crop growth processes, potentially could make a large contribution to furthering our understanding of fertilizer behavior in cropping systems. Such models, with the capability of readily simulating various crop and fertilizer management strategies, should lead to a great improvement in the efficacy of fertilizer decision making. Optimizing fertilization strategies, given the uncertainties of climate, is generally difficult. The problem is compounded in some less-developed regions of the world where fertilizer data are sparse. Where adequate climatic, soil, and crop data exist, simulation models will allow some extrapolation into these less-developed areas and, thus, provide some insights into fertilizer behavior in various environments.

Many different simulation models exist that describe some or all of the N-cycle processes occurring in cropping systems. Some of these were the subject of the proceedings of a workshop (Frissel & van Veen, 1981), and of reviews by Tanji (1982). Several of the models cited in these reviews are concerned with specific aspects of the N cycle, such as ammonia volatilization (Parton et al., 1981), leaching (Addiscott, 1981; Burns, 1980), and denitrification (Smith, 1981; Leffelaar, 1981). Most of the models are primarily concerned with the major soil processes in the N cycle and few consider the N dynamics of a growing crop. Several very comprehensive models have also been developed that contain details that link together all the major soil transformations. Among them, those of van Veen (1977) and Molina et al. (1983) provide considerable detail about the functioning of the various transformations at the microbiological level.

Other models have been developed with the specific purpose of examining aspects of pollution from organic wastes and from excessive fertilization (Rao et al., 1981; Selim & Iskandar, 1981; Donigian & Crawford, 1976). Some models, such as those of Watts and Hanks (1978), Tillotson et al. (1980), and Tanji et al. (1981), simulate most of the major soil transformations of N as well as the uptake by the crop, but fall short of fully describing the system in that the crop growth and yield response is not incorporated. A simulation model that predicts dry-matter production as limited by water and N, PAPRAN (Seligman & van Keulen, 1981) is used to make predictions for pastures in semiarid environments.

Many of these models were designed for a specific purpose and do not lend themselves to integration with a general crop and water balance model. To simulate N dynamics adequately in a range of diverse cropping environments, a model capable of describing the major soil N transformations, as well as the plant component, is required. The CERES-N model described below is the nitrogen component of the CERES-WHEAT (Jones & Kiniry, 1986) and CERES-MAIZE (Godwin & Vlek, 1985) models. The CERES models simulate growth, phenology, water and N balance, and yield. They were designed to have widespread applicability in diverse environments and to require only a minimum data set of commonly available field data as inputs.

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I. THE CERES-N MODEL

The N component of the model is not designed to operate in a stand-alone mode, but as a component of the CERES models. Complete documentation of the CERES models has been published elsewhere (for maize [*Zea mays* L.] see Jone & Kiniry, 1986, and for wheat [*Triticum aestivum* L.] see Ritchie et al., 1988). The CERES models can be run with an option set to simulate the condition in which N is not limiting. This version describes evapotranspiration, soil water balance, and crop development as influenced by temperature, vernalization (wheat only) and photoperiod, vegetative growth, root growth, and grain growth. The N components add to this the description of mineralization and/or immobilization of N associated with the decay of crop residues, nitrification, denitrification, urea hydrolysis, leaching of nitrate, and the uptake and use of N by the crop. The N model uses the layered soil-water balance model described by Ritchie (1985) and the soil temperature component of the EPIC model (Williams et al., 1983). Both these submodels are integrated into the CERES models.

The N model comprises four subroutines describing N movement, soil N transformations, N uptake, and plant N stress factors. A listing of these subroutines can be found in Appendix 13-1 and a glossary of the variables is listed in Appendix 13-2. The effects of N deficiency on plant growth processes are incorporated into the main growth subroutine (GROSUB) of the CERES models. Additional subroutines are used within the CERES models for input and output of data, and for the initialization of the various N pools. Since the model is not designed to operate in a stand-alone manner, it is difficult to estimate the computer time required for execution of the N components. By comparing CERES model runs with the option set to simulate or not simulate the N balance, an approximate estimate can be obtained. On a VAX 11/750 computer, simulation of crop growth and water balance for one complete season requires approximately 7 s of CPU time. The additional simulation of the N balance requires approximately 3 s. On an IBM PC AT equipped with floating point coprocessor, corresponding times are 25 and 33 s, respectively.

A. Nitrate and Urea Movement

Leaching of nitrate (NO_3) is probably the most common and the best understood N loss process. Nitrates leaching from soil often become a source of contamination of ground water. This has generated more recent interest in leaching from an environmental standpoint (Wild & Cameron, 1980).

The many approaches to modeling leaching was the subject of a review by Addiscott and Wagenet (1985). Many of these approaches are based on numerical techniques, which require solution in a manner inappropriate for use in a management-level model such as CERES. In the CERES model, only the movement of nitrate and urea is considered. Ammonium is assumed not to be transported across soil layers. The same procedures for simulating nitrate movement are used for simulating urea movement.

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Nitrate movement in the soil profile is dependent upon water movement. In the water balance component of the model the volume of water moving from a layer (L) to the layer below [FLUX(L)] is calculated. The volume of water present in the layer before drainage occurred is also calculated from the volumetric water content [SW(L)] and the depth of the layer [DLAYR(L)]. Nitrate lost from each layer (NOUT) may then be calculated as a function of the water that is retained and the water that moves as

$$\text{NOUT} = \text{SNO3(L)} \times \text{FLUX(L)}/[\text{SW(L)} \times \text{DLAYR(L)} + \text{FLUX(L)}] \quad [1]$$

where SNO3(L) is the quantity of nitrate present in layer L (kg N/ha). Thus, a fraction of the mass of nitrate [SNO3(L)] present in each layer moves with each drainage event. A simple cascading approach is used in which the nitrate lost from one layer is added to the layer below. When the concentration of nitrate in a layer falls to 1.0 μg NO_3/g of soil, then no further leaching from that layer is allowed to occur. The implicit assumption is that all the nitrate present in a layer is uniformly and instantaneously in solution in all of the water in the layer.

Nitrate is more readily displaced from sands since the volume of water that can move is large in comparison with the retained water. Most of the difference in the simulated leaching rate between soils of different texture is explained by this difference in the proportion of water that is mobile. Some difference is also attributable to the rate at which the profile can drain.

Similar procedures are used to model the rate of upward movement of nitrate and urea with evaporation of water from the surface layers. In this case, the water balance routine calculates the upward flow of water [FLOW(L)] and the amount of upward movement (NUP) is calculated as for NOUT as

$$\text{NUP} = \text{SNO3(L)} \times \text{FLOW(L)}/[\text{SW(L)} \times \text{DLAYR(L)} \times \text{FLOW(L)}] \quad [2]$$

This movement of N is confined to the upper layers of the soil profile and is generally small, due to the small volume of moving water. No upwards loss from the top layer occurs by this process.

B. Soil Nitrogen Transformations

In subroutine NTRANS, the CERES model simulates the decay of organic matter and the subsequent mineralization and/or immobilization of N, the nitrification of ammonium, and denitrification. Fertilizer addition and urea hydrolysis are also performed in this subroutine.

C. Fertilizer Additions and Urea Hydrolysis

Fertilizer N is partitioned in the model between nitrate, ammonium, and urea pools according to the nature of the fertilizer used. The assumption is

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made that the fertilizer is uniformly incorporated into the soil layer into which it is placed. Surface fertilizer applications are treated as being uniformly incorporated into the top layer. Up to 10 split applications can be accommodated by the model.

To simulate urea hydrolysis, a maximum hydrolysis rate is estimated from the soil organic carbon and pH. Data to estimate this rate were drawn from several laboratory studies (Tabatabai & Bremner, 1972; McGarity & Myers, 1967; Myers & McGarity, 1968; Zantua et al., 1977; Beri & Brar, 1978). This maximum hydrolysis rate is then scaled downward according to the prevailing moisture and temperature conditions. The functions used are:

$$AK = -1.12 + 1.31 \times OC(L) + 0.203 \times PH(L) - 0.155 \times OC(L) \times PH(L) \quad [3]$$

where AK is the maximum hydrolysis rate (dimensionless rate constant), OC(L) is the organic carbon in layer L (%), PH(L) = soil pH in layer L, and AK has an assumed minimum value of 0.25. The actual hydrolysis rate (UHYDR) is estimated as:

$$UHYDR = AK \times AMIN1(SWF,TF) \times UREA(L) \quad [4]$$

where SWF,TF represents zero to unity indices for soil water and temperature, respectively, UREA(L) is the quantity of urea present in a layer (kg urea/ha), and AMIN1 is a FORTRAN function in which the minimum of the arguments is selected. The temperature and soil water indices are designed to simulate the effects of soil moisture and temperature on urea hydrolysis reported in a laboratory study by Vlek and Carter (1983). Any urea remaining after 21 d is assumed to have hydrolyzed. The N released by the hydrolysis of urea is added to the soil ammonium pool.

D. Mineralization and Immobilization

Mineralization refers to the net release of mineral N with the decay of organic matter, and immobilization refers to the transformation of inorganic compounds to the organic state. Both processes are microbial in origin. Immobilization occurs when soil microorganisms assimilate inorganic N compounds and use them in the synthesis of the organic constituents of their cells. A balance exists between the two processes. When crop residues with a high C:N ratio are added to soil, the balance can shift, resulting in net immobilization for a period of time. After some of the soil carbon has been consumed by respiration, net mineralization may resume. Nitrogen mineralized from the soil organic pool often constitutes a large part of the N available to the crop.

The perceived application for the CERES models in studies examining crop growth and fertilizer management requires that a mineralization model be simple, require few inputs, and able to work on a diversity of soils. Simulation studies examining the effects of crop residues also require that the

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model be capable of simulating the fate of residues of various compositions. Other studies examining the potential role of nitrification inhibitors require a model wherein the processes of ammonification and nitrification are separated. The approach used in the CERES-WHEAT model is based on a modified version of the mineralization and immobilization component of the PAPRAN model (Seligman & van Keulen, 1981). This model attempts to maintain some of the functionality of the microbiological-level models, but does so at a very simplified level. The modifications to the model have been performed to first simulate nitrification rather than assume it as an instantaneous process. Secondly, the fresh organic matter pools simulated were partitioned further, so that an interface to the denitrification procedures could be constructed. Modifications were also made to temperature and water indices to fit the CERES water balance and soil temperature routines. Unless indicated otherwise, the coefficients used for the mineralization/immobilization functions described below were drawn from the PAPRAN model. The mineralization and immobilization routine simulates the decay of two types of organic matter. These are fresh organic matter (FOM), which includes crop residues or green manure, and a more stable organic or humic pool (HUM). The FOM pool is further subdivided into three pools: carbohydrate, cellulose, and lignin.

In PAPRAN, FOM is simulated as one pool and the decay rate constant is selected according to the proportion of the initial amount of FOM remaining. In CERES, separation of FOM into three pools enabled a better estimate of soluble carbon, which is used in the denitrification routine.

Initially, the FOM(L) contains 20% carbohydrate, 70% cellulose, and 10% lignin. The model requires, as input data , the amount of straw added, its C/N ratio, and its depth of incorporation (if any). An estimate of the amount of root residue from the previous crop is also required. These data are used to initialize FOM and the N contained within (FON) for each layer. The FOM is measured in units of kilograms of organic matter per hectare and FON is measured in kilograms of N per hectare. The mineralization routine also requires the soil organic carbon in each layer [OC(L)] as an input. This is used to calculate HUM(L) and, together with a simplifying assumption of bulk soil C/N ratio of 10, is used to estimate the N associated with this fraction [NHUM(L)].

Each of the three FOM pools has a different decay rate [RDECR(1 to 3)]. Under nonlimiting conditions, the decay constants as reported by Seligman and van Keulen (1981) are 0.80, 0.05, and 0.0095 for carbohydrate, cellulose, and lignin, respectively. The decay constant for carbohydrate implies that, under nonlimiting conditions, 80% of the pool will decay in 1 d. Nonlimiting conditions very seldom occur in soils since one or all of soil temperature, soil moisture, or residue composition will limit the decay process. To quantify these limits, three (zero to unity) factors are calculated. A water factor (MF) is first determined from the volumetric soil water content [SW(L)] relative to the lower limit (LL) and drained upper limit (DUL) (Fig. 13–1).

The function for MF follows the observations reported by Myers et al. (1982) and Linn and Doran (1984) on moisture effects on ammonification.

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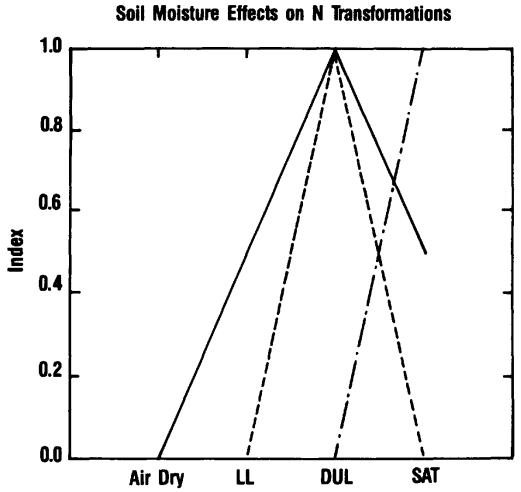


Fig. 13-1. Soil water indices used to modify soil N transformation rates (LL = lower limit moisture content, DUL = drained upper limit moisture content, SAT = saturation moisture content, —— = ammonification, --- = nitrification, —·— = denitrification).

Under very wet conditions (100% of water filled porosity), ammonification proceeds at approximately half of the rate of ammonification at field capacity (Linn & Doran, 1984).

The temperature factor (TF) approximates the soil temperature effects on ammonification reported by others (Stanford et al., 1973; Myers, 1975). The C/N ratio (CNR) imposes the third limit on decay rate. In this case, CNR is calculated as the carbon contained in FOM divided by the N available for the decay process. This N available for decay is the sum of the N contained in the FOM and the extractable mineral N present in the layer. From CNR, an index (CNRF) is calculated, which has a critical CNR of 25 (Fig. 13-2). In low N containing residues (e.g., freshly incorporated wheat straw) with a high CNR, the N available for the decay process will greatly limit the decay rate.

For each of the FOM pools, a decay rate appropriate for that pool (JP) can be calculated by multiplying the rate constant by the three indices as

$$G1 = TF \times MF \times CNRF \times RDECR(JP) \quad [5]$$

where G1 is then the proportion of the pool that decays in 1 d. The amount of material that has decayed is then the product of G1 and the pool size. The gross mineralization of N associated with this decay (GRNOM) is then calculated according to the proportion of the pool that is decaying as

$$GRNOM = G1 \times FPOOL(L,JP)/FOM(L) \times FON(L) \quad [6]$$

where FPOOL(L,JP) is the pool of either carbohydrate (JP = 1), cellulose (JP = 2), or lignin (JP = 3) present in layer L (μg/ha). The GRNOM is summed for each of three pools in each layer. Similarly, GRCOM, the amount of decaying organic matter, is determined as the sum of three pool fractions.

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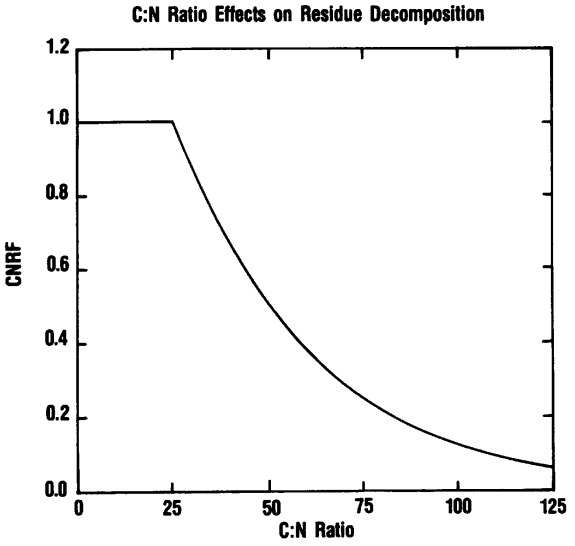


Fig. 13-2. Effect of C/N ratio on the index (CNRF) for residue decay rate.

The procedure used for calculating the N released from the humus (RHMIN) also uses TF and MF. In this case, CNRF is not used and the potential decay rate constant (DMINR) is very small (8.3×10^{-5}). A further index (DMOD) was added to the RHMIN calculations to adjust the mineralization rate for certain atypical soils. On certain volcanic ash soils mineralization proceeds slowly, necessitating a less than unity value of DMOD so that mineralization is not overestimated. On freshly cultivated virgin soils, a slightly greater than unity value has been found necessary to account for the sudden increase in mineralization activity. In all other circumstances a value of 1.0 is used for DMOD. Satisfactory alternatives for estimating DMOD are currently being sought. The procedure for calculating RHMIN, then, is the product of the various indices and the N contained within the humus [NHUM(L)].

$$\text{RHMIN} = \text{NHUM(L)} \times \text{DMINR} \times \text{TF} \times \text{MF} \times \text{DMOD} \quad [7]$$

These calculations also allow for the transfer of 20% of the gross amount of N released by the mineralization of FON(L) to be incorporated into NHUM(L). This accounts for N incorporated into microbial biomass (Seligman & van Keulen, 1981).

As organic matter decomposes, some N is required by the decay process and may be incorporated into microbial biomass. The N that is immobilized in this way (RNAC) is calculated as the minimum of the soil-extractable mineral N (TOTN) and the demand for N by the decaying FOM(L) as

$$\text{RNAC} = \text{AMIN1(TOTN,GRNOM)} \times [0.02 - \text{FON(L)/FOM(L)}] \quad [8]$$

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where 0.02 is the N requirement for microbial decay of a unit of FOM(L). The value of 0.02 is the product of the fraction of carbon in the FOM(L) (40%), the biological efficiency of carbon turnover by the microbes (40%), and the N/C ratio of the microbes (0.125). The FOM(L) and FON(L) are then updated as

$$\text{FOM(L)} = \text{FOM(L)} - \text{GRCOM} \quad [9]$$

$$\text{FON(L)} = \text{FON(L)} + \text{RNAC} - \text{GRNOM} \quad [10]$$

The balance between RNAC and GRNOM determines whether net mineralization or immobilization occurs. The net N released from all organic sources (NNOM) is

$$\text{NNOM} = 0.8 \times \text{GRNOM} + \text{RHMIN} - \text{RNAC}. \quad [11]$$

Note that only 80% of GRNOM enters this pool since the remaining 20% was incorporated into NHUM(L). The NNOM can then be used to update the ammonium pool [SNH4(L)].

$$\text{SNH4(L)} = \text{SNH4(L)} + \text{NNOM}. \quad [12]$$

If net immobilization occurs (NNOM negative) ammonium is first immobilized and, if there is not sufficient ammonium to retain this pool with a concentration of 1 ppm, then withdrawals are made from the nitrate pool.

E. Nitrification

Nitrification refers to the process of oxidation of ammonium to nitrate. It is a biological process and occurs under aerobic conditions. The main factors that limit nitrification are substrate NH_4^+, oxygen, soil pH, and temperature (Focht & Verstraete, 1977).

The approach used in the CERES models has been to calculate a potential nitrification rate and a series of zero to unity environmental indices to reduce this rate. The potential nitrification rate is a Michaelis-Menten kinetic function dependent only on ammonium concentration and thus, is independent of soil type. A further index, termed a nitrification capacity index, is introduced, which is designed to introduce a lag effect on nitrification if conditions in the immediate past (previous 2 d) have been unfavorable for nitrification. Actual nitrification capacity is calculated by reducing the potential rate by the most limiting of the environmental indices and the capacity index. The functions reported below were found to be appropriate across the range of data sets tested.

The key functions are

$$\text{A} = \text{AMIN1(RP2,WFD,TF,PHN)} \quad [13]$$

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where RP2 is the zero to unity index for nitrification potential based on the immediate past (unitless), WFD is the zero to unity index for the soil moisture effect on nitrification rate (unitless), TF is the zero to unity index for soil temperature effects on nitrification rate, and PHN is the zero to unity function for the soil pH effect on nitrification rate. A Michaelis-Menten function described by McLaren (1970) was modified to incorporate these environmental indices and to determine the proportion of the soil ammonium pool that nitrifies in 1 d.

$$\text{RNTRF} = [\text{A} \times 40.0 \times \text{NH}_4(\text{L})]/[\text{NH}_4(\text{L}) + 90.0] \times \text{SNH}_4(\text{L}) \quad [14]$$

where RNTRF is the rate of nitrification (kg N ha^{-1} d^{-1}), $NH_4(L)$ is the concentration of ammonium N in layer L (μg N/g soil), and $SNH_4(L)$ is the amount of ammonium in layer L (kg N/ha).

F. Denitrification

Denitrification is the dissimilatory reduction of nitrate (or nitrite) to gaseous products including NO, N_2O, and N_2 (Knowles, 1981). Denitrification is a microbial process that occurs under anaerobic conditions and is influenced by organic carbon content, soil aeration, temperature, and soil pH (Focht & Verstraete, 1977).

The approach adopted in the CERES models has been to adapt the functions described by Rolston et al. (1980) to fit within the framework of the model, and to match inputs derived from the water balance and mineralization components of CERES. The basic function used by these authors was also used by Davidson et al. (1978) and was the subject of field testing under a variety of conditions in California. Predicted rates of denitrification compared favorably with direct measures of gaseous losses in the field experiments.

Denitrification calculations are performed only when the soil water content (SW) exceeds the drained upper limit (DUL). A zero to unity index (FW) (see Fig. 13-1) for soil water in the range from DUL to saturation (SAT) is calculated as

$$\text{FW} = 1.0 - [\text{SAT(L)} - \text{SW(L)}]/(\text{SAT(L)} - \text{DUL(L)}] \quad [15]$$

Linn and Doran (1984) used percentage of water-filled porosity as an index of soil-water availability effects on soil N transformations. In their studies, denitrification commenced with a water-filled porosity of 60% and increased linearly up to 100% water-filled porosity. This approximates the linear increase in FW as SW increases from DUL to SAT.

A factor for soil temperature (FT) is also calculated.

$$\text{FT} = 0.1 \times \exp[0.046 \times \text{ST(L)}] \quad [16]$$

where ST(L) is the temperature (°C) of layer L. Rolston et al. (1980) estimate the water-extractable carbon in soil organic matter (CW) as

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$$CW = 24.5 + 0.0031 \times SOILC \quad [17]$$

In the CERES model, SOILC is calculated as 58% of the stable humic fraction. To this is added the carbon contained in the carbohydrate fraction of the FOM pool. Denitrification rate (DNRATE) is then calculated from the nitrate concentration and converted to a kg N/ha basis for the mass balance calculations.

$$DNRATE = 6.0 \times 1.0 \times 10^{-5} \times CW \times NO_3(L) \times FW \times FT \times DLAYR(R)$$

where FT is the temperature factor effect on denitrification (unitless), FW is the water factor effect on denitrification (unitless), $NO_3(L)$ is the nitrate concentration in layer L (μg N/g soil), CW is the total water extractable carbon in the soil layer (μg carbon/g soil), and DNRATE is the denitrification rate (kg N ha^{-1} d^{-1}).

G. Plant Nitrogen Concentrations

Plant growth is greatly affected by the supply of N. Typically the supply of N to plants at the beginning of the season is relatively high and becomes lower as the plant reaches maturity (Jones, 1983). The concentration of N in plant tissues also changes as the plant ages. During early growth, N concentrations are usually high due to synthesis of large amounts of organic N compounds required by the biochemical processes constituting photosynthesis and growth (Novoa & Loomis, 1981). As the plant ages, less of this new material is required and export from the old tissue to new tissue occurs, lowering the whole-plant N concentration. At any point in time there exists a critical N concentration in the plant tissue below which growth will be reduced. These concentrations are determined as a function of crop ontogenetic age and are used within the model as part of the procedure to simulate the effects of N deficiency. Critical N concentration functions related to plant age have been determined for maize (Jones, 1983) and for wheat (D.C. Godwin, 1987, unpublished data). These authors also defined critical concentrations for plant roots for these two crops, and minimum concentrations for tops and roots in both crops.

The critical and minimum concentrations are used to define a N factor (NFAC), which ranges from zero to slightly above unit. The NFAC is the primary mechanism used within the model to determine the effect of N on plant growth. It is an index of deficiency relating the actual concentration in tops (TANC) to these critical concentrations. The NFAC has a value of zero when TANC is at its minimum value of TMNC and increases to 1.0 as concentration increases toward the critical concentration. It is calculated as

$$NFAC = 1.0 - (TCNP - TANC)/(TCNP - TMNC) \quad [18]$$

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where TCNP is the tops critical N concentration (g N/g DW). Since all plant growth processes are not equally affected by N stress, a series of indices based on NFAC are used. The shape and nature of these differ between the maize and wheat models. For the calculation of these indices, NFAC has a maximum value of 1.0. This implies that when TANC exceeds TCNP no extra growth occurs. The relationship between these deficiency indices and NFAC is depicted in Fig. 13-3.

The first of these indices (NDEF1) is used to describe the effects of N deficiency on photosynthesis per unit leaf area. The second (NDEF2) describes the effect of N deficiency on the rate of leaf area expansion and leaf senescence. In the wheat model a third index (NDEF3) is used to describe the effect of N deficiency on tillering, and a fourth factor (NDEF4) is used to modify the rate of grain N accumulation. In the growth component of the models, these N deficiency indices, together with similarly defined zero to unity indices for soil water stress, are used to reduce potential rates of the growth processes. The more limiting of either soil water or N stress is selected in each case to modify the process.

H. Nitrogen Uptake

The approach used in the CERES models calculates the components of crop demand for N and the soil supply of N separately, and then uses the lesser of these two to determine the actual rate of uptake. Demand can be considered to have two components. First, there is a deficiency demand. This is the amount of N required to restore the actual N concentration in the plant (TANC for tops) to the critical concentration (TCNP for tops). Critical concentrations for shoots and roots were previously defined. This deficiency demand can be quantified as the product of the existing biomass and the concentration difference as

$$\text{TNDEM} = \text{TOPWT} \times (\text{TCNP} - \text{TANC}) \quad [19]$$

where TOPWT is the weight of plant tops (kg/ha) and TNDEM is the plant tops N demand (kg N/ha). Similary, for roots the discrepancy in concentration (difference between RCNP and RANC) is multiplied by the root biomass (RTWT) to calculate the root N demand

$$\text{RNDEM} = \text{RTWT} \times (\text{RCNP} - \text{RANC}). \quad [20]$$

Luxury consumption of N can occur when the supply of N to the plant is large and the growth of the plant is slowed due to some environmental washout, such as low light flux density. If luxury consumption of N has occurred such that TANC is greater than TCNP, then these demand components have negative values. If total N demand is negative, then no uptake is performed on that day. As biomass increases with crop growth, plant N tends to become diluted and the concentration falls. Thus, when TANC is greater than TCNP, a period of growth will generally cause TANC to fall toward or below TCNP.

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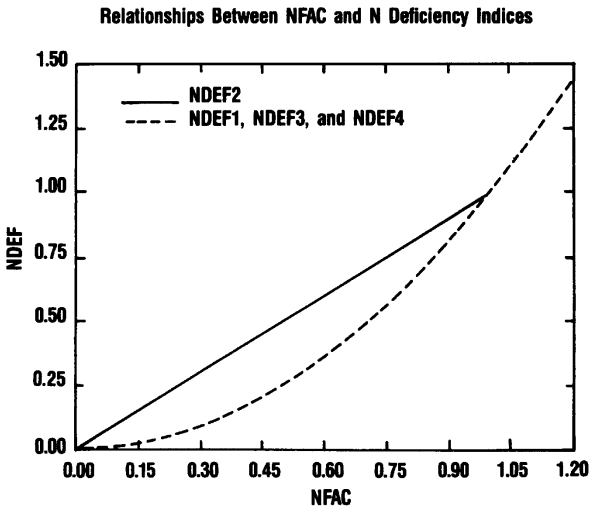


Fig. 13-3. Relationship between N deficiency index (NFAC) and the N deficit factors (NDEF) used to modify plant growth processes.

The second component of N demand is that needed by the new growth. Here the assumption is made that the plant would attempt to maintain a critical N concentration in the newly formed tissues. To calculate the new growth demand a potential amount of new growth is first estimated in the GROSUB subroutine. New growth is estimated from potential photosynthesis (PCARB) and is partitioned into a potential root growth (PGRORT) and a potential tops growth (PDWI). These potential growth increments provide a mechanism for the tops actual N concentration (TANC) to exceed TCNP. This occurs when some stress prevails and the actual growth increment is less than the potential. New growth demand for tops (DNG) is calculated as

$$DNG = PDWI \times TCNP \quad [21]$$

and the new growth demand for roots is calculated as the product of

$$PGRORT \times RCNP. \quad [22]$$

During the early stages of plant growth, the new growth component of N demand will be a large proportion of the total demand. As the crop biomass increases, the deficiency demand becomes the larger component. During grain filling, the N required by the grain is removed from the vegetative and root pools to form a grain N pool. The resultant lowering of concentration in these pools may lead to increased demand. The total plant N demand (ANDEM) is the sum of all of these demand components.

To calculate the potential supply of N to the crop, zero to unity availability factors for nitrate (FNO3) and ammonium (FNH4) are calculated from the soil concentrations of the respective ions as

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$$FNO3 = 1.0 - \exp[-0.0275 \times NO3(L)] \quad [23]$$

$$FNH4 = 1.0 - \exp[-0.025 \times NH4(L)] \quad [24]$$

The coefficients used in these two functions, obtained by trial and error, were found to be appropriate for several data sets from the literature.

A zero to unity soil-water factor (SMDFR), which reduces potential uptake, is calculated as a function of the relative availability of soil water

$$SMDFR = [SW(L) - lL(L)/ESW(L) \quad [25]$$

where ESW(L) is the extractable soil water in the layer L. To account for increased anaerobiosis and declining root function at moisture contents above the drained upper limit, SMDFR is reduced as saturation is approached. The maximum potential N uptake from a layer may be calculated as a function of the maximum daily uptake per unit length of root and the total amount of root present in the layer. The first of these is a temporary variable (RFAC), which integrates the effects of root length density [RLV(L)] (cm root/cm^3 soil), the soil water factor described above, and the depth of the layer [DLAYR(L) (cm) as

$$RFAC = RLV(L) \times SMDFR \times SMDFR \times DLAYR(L) \times 100.0 \quad [26]$$

The second of these equations incorporates the ion concentration effect (FNO3) and the maximum daily uptake per unit length of root (0.006 kg N/ha cm root) to yield a potential uptake of nitrate from the layer [RNO3U(L)].

$$RNO3U(L) = RFAC \times FNO3 \times 0.006. \quad [27]$$

Thus, [RNO3U(L)] is the potential uptake of nitrate from layer L in kg N/ha constrained by the availability of water, the root length density, and the concentration of nitrate. Initial estimates for the maximum uptake per unit length of root coefficient were obtained from the maize root data of Warncke and Barber (1974). This estimate was the subject of continuing modification during early model development. The value reported here appears to be appropriate across a broad range of data sets. The effect of each of these parameters on determining potential uptake can be seen in Fig. 13-4. A similar function is employed to calculate the potential uptake of ammonium [RNH4U(L)].

$$RNH4U(L) = RFAC \times FNH4 \times 0.006 \quad [28]$$

To account for declining root function with increasing plant age during grain filling, a further term to reduce RFAC is also introduced.

Potential N uptake from the whole profile (TRNU) is the sum of RNO3U(L) and RNH4U(L) from all soil layers where roots occur. Thus, TRNU represents an integrated value that is sensitive to (i) rooting density;

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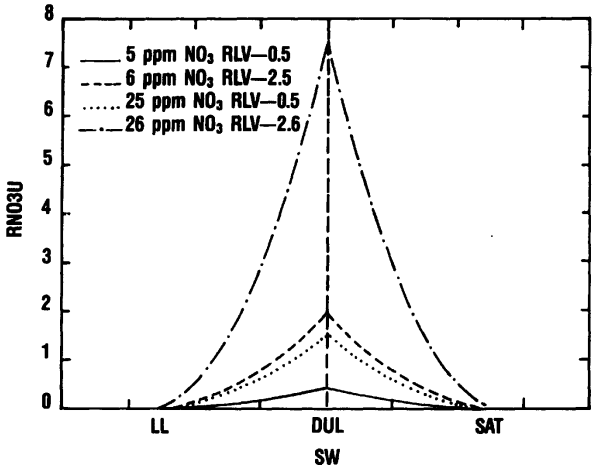


Fig. 13-4. Simulated effect of changing soil water status (SW) on uptake of nitrate from a layer (RNO3U). The LL, DUL, SAT represent the lower limit, drained upper limit, and saturation moisture contents, respectively. The four curves refer to differing concentrations of nitrate (ppm NO_3) and differing root length densities (RLV in cm root/cubic cm of soil).

(ii) the concentration of the two ionic species; and (iii) their ease of extraction as a function of the soil-water status of the different layers. This method of determining potential uptake enables the condition of nutritional drought to be simulated. Nutritional drought occurs when nutrients and roots are concentrated in the upper layers of the soil profile, but sufficient water for growth and uptake is present only in the lower layers.

If the potential N supply from the whole profile (TRNU) is greater than the crop N demand (ANDEM), a N uptake factor (NUF) is calculated and used to reduce the N uptake from each layer to the level of demand.

$$\text{NUF} = \text{ANDEM}/\text{TRNU}. \qquad [29]$$

This could occur when plants are young and have a high N supply. If the demand is greater than the supply, then NUF has a value of 1.0. When NUF is <1.0, uptake from each layer is reduced.

Under conditions of luxury N upake (TANC > TCNP) exudation of organic N compounds can occur. Rovira (1969) found that changes in the shoot environment, which cause more rapid growth, can increase exudation and Bowen (1969) reported that N deficiency can cause it to decrease. In the CERES-N model this exuded N is added to the fresh organic N pool [FON(L)], and can be mineralized and subsequently made available to the plant again. The amount of N that can be lost daily from the plant in this manner is calculated as 5% of the N contained in the roots/day.

Following uptake, concentrations of N in each of the shoots and roots are updated. Partitioning of the N taken up between shoot and root parts occurs on the basis of the proportions of the total plant demand arising from shoots and roots, respectively.

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I. Grain Nitrogen Concentration

In many crop-growing areas, when the crop reaches the grain-filling stage soil supplies of N are very low. In these cases the N requirement of the developing grains is largely satisfied by remobilization of protein N from vegetative organs. When N supply is increased, the proportion of grain N arising from remobilization declines and the proportion from uptake increases (Vos, 1981). Many studies (Benzian et al., 1983, Terman et al., 1969) have found negative correlations between grain yield and grain protein concentration. Temperature and soil moisture also affect the grain N content.

Grain N determination is simulated differently in the maize and wheat models. In maize, the concentration of N in each day's new grain growth is modeled as a function of a temperature index, a soil water index, and an index based on NFAC. This is accomplished as follows.

First, two factors, TFAC and SFAC, are calculated and used to estimate the effects of mean temperature (TEMPMN) and drought stress (SWDF2) on the concentration of nitrogen (GNP) in GROGRN for the day. The AMAX1 is a FORTRAN function subprogram that finds the maximum of two or more values.

$$\text{TFAC} = 0.69 + 0.125 \times \text{TEMPM} \quad [30]$$

$$\text{SFAC} = 1.125 - 0.125 \times \text{SWDF2} \quad [31]$$

$$\text{GNP} = (0.007 \times 0.010 \times \text{NDEF2}) \times \text{AMAX1(TFAC,SFAC)} \quad [32]$$

Thus, high mean temperatures and drought stress can cause the value of AMAX1(TFAC,SFAC) to exceed 1.0. Nitrogen deficiency can cause NDEF2 (calculated in subroutine NFAC) to be <1.0. The net effect of these equations is to allow GNP to range from <0.010 when N deficiency is severe to about 0.018 when adequate N is available. However, high temperatures or drought stress limit grain growth.

In the wheat model the N accumulation rate of single grains is simulated as a function dependent on the daily maximum and minimum temperature and is quite similar to the functions for the rate of grain mass accumulation. Grain N accumulation is more sensitive to temperature than grain mass accumulation, thus, at high temperatures grain N concentrations will tend to be higher. Moisture stress affects crop photosynthesis, which supplies assimilate to developing grains. Moisture stress will affect the simulated grain N concentration in wheat by altering the dilution of N in the grain by assimilate.

In both models, a daily grain N sink is determined based on this daily concentration of N in new grain growth and the amount of new grain growth. In the wheat model, the sink is adjusted according to the N status of the plant. This will influence grain N concentration.

Two pools of N within the plant are available for translocation: a shoot pool (NPOOL1) and a root pool (NPOOL2). These pools are determined

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from the N concentration (VANC or RANC) relative to the critical concentration (VMNC or RMNC) and the biomass of the pool (RTWT or TOPWT).

$$\text{NPOOL1} = \text{TOPWT} \times (\text{VANC} - \text{VMNC}) \quad [33]$$

$$\text{NPOOL2} = \text{RTWT} \times (\text{RANC} - \text{RMNC}) \quad [34]$$

The total N available for translocation (NPOOL) is the sum of these two pools. When NPOOL is not sufficient to supply the grain N demand (NSINK), NSINK is reduced to NPOOL. If NSINK is greater than that which can be supplied by the tops (NPOOL1), then NPOOL1 is set to zero and the tops N concentration is set to its minimum value (VMNC). The remaining NSINK is then satisfied from the root N pool and the N pool is updated accordingly.

$$\text{NPOOL2} = \text{NPOOL2} - (\text{NSINK} - \text{NPOOL1}) \quad [35]$$

The root N concentration is then updated.

$$\text{RPOOLN} = \text{RTWT} \times \text{RMNC} \times \text{NPOOL2} \quad [36]$$

$$\text{RANC} = \text{ROOTN/RTWT} \quad [37]$$

When NSINk < NPOOL1, it is totally satisfied from tops N

$$\text{NPOOL1} = \text{NPOOL1} - \text{NSINK} \quad [38]$$

and the tops concentrations are updated accordingly.

$$\text{TOPSN} = \text{NPOOL1} + \text{VMNC} \times \text{TOPWT} \quad [39]$$

$$\text{VANC} = \text{TOPSN/TOPWT} \quad [40]$$

The total amount of N contained in the grain can tehn be accumulated

$$\text{GRAINN} = \text{GRAINN} + \text{NSINK}. \quad [41]$$

This routine, together with the remainder of the growth routine and the N deficiency indices, thus provides several pathways by which N stress during grain filling can affect grain yield and grain protein content. First, as N is removed from the vegetative tissues, NFAC will become lower. This will, in turn, lower NDEF4 and lower the sink size for N, thus providing for the capability of reduced grain N concentration. Lowering NFAC will also lower NDEF1, which will cause the rate of crop photosynthesis to fall, thus lowering the assimilate available for grain filling. A declining NFAC will also speed the rate of senescence, which will reduce the leaf area available for photosynthesis. Soil water stress during grain filling can also increase

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the grain N concentration since it will reduce photosynthesis. This would lower assimilate availability and, thus, not dilute grain N as much as would occur in an unstressed crop.

II. MODEL INPUTS

To facilitate model usage across a wide range of locations and model applications, the required input data have been kept to a minimum. In addition to the data required by the CERES crop growth models, the N model requires extractable soil nitrate and ammonium concentrations, organic carbon, bulk density, and pH in each layer. The model also requires an estimate of the amount of crop residue present and its C/N ratio, as well as fertilizer information including the date, amount, and depth of all applications made to the crop, and the type of fertilizer used.

III. MODEL EVALUATION

Testing of the N component of the CERES models has occurred simultaneously with the testing of the crop growth components of the model. Detailed documentation of this testing can be found in Otter-Nacke et al. (1985) and Godwin (1987, unpublished data) for the wheat model and in Jones and Kiniry (1986) and Singh (1985, unpublished data) for the maize model. In the case of both models, test data sets for these two crops were obtained from diverse locations spanning the world's cropping regions. These data sets also represented a diversity of soil types, fertilizer practices, and climatic conditions. In each data set, pests, diseases, and nutrients other than N were not considered to be limiting to growth. The testing of each model (Fig. 13–5, 13–6, and 13–7) indicates that the model generally provides reasonable predictions of grain-yield N uptake by the aboveground plant and the partitioning of this N into grain.

IV. MODEL LIMITATIONS AND USES

It should be stressed that the model can only account for variations in the factors defined in the model's description and this assumes that other potentially important factors, such as other nutrients, pests, and diseases, are nonlimiting. The model is unable to account for volatile losses of ammonia from the soil surface, leaching of ammonium N, or direct losses of N from plants into the atmosphere. Where these losses are substantial, the model will be inaccurate. Since the soil water and N components of the model

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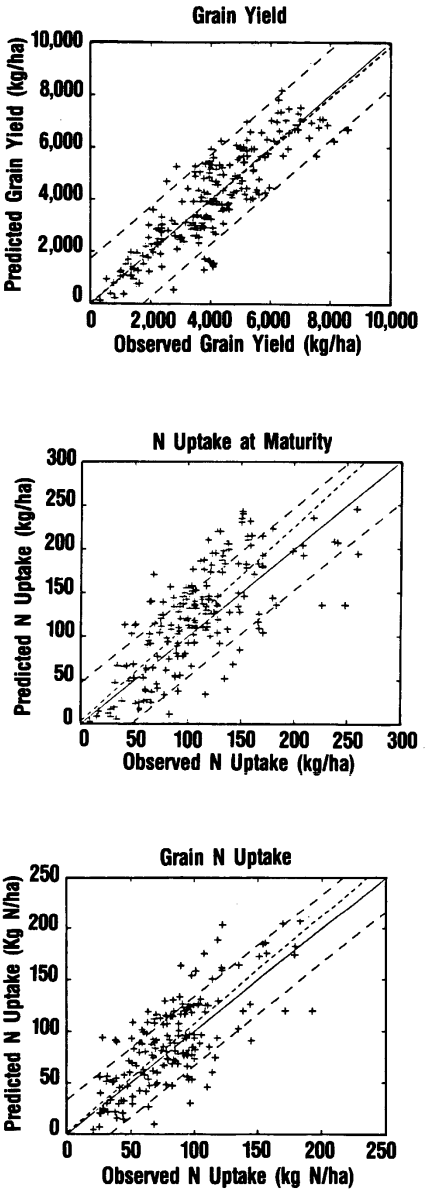


Fig. 13-5. Comparison of predictions of the CERES-WHEAT model with observed data from experiments: (a) Grain yield; (b) Nitrogen uptake at maturity; and (c) Grain N uptake (from Otter-Nacke et al., 1985).

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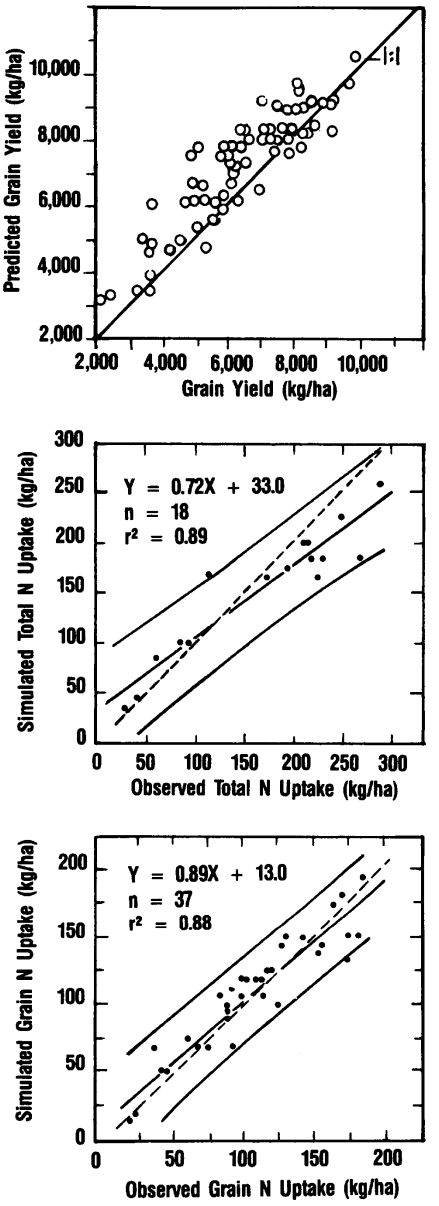


Fig. 13-6. Comparison of predictions of the CERES-MAIZE model with observed data from field experiments: (a) Grain yield (from Singh, 1985, unpublished data); (b) Crop N uptake at maturity for four cultivars at four locations (from Jones and Kiniry, 1986); and (c) Grain N uptake at maturity for nine cultivars at seven sites.

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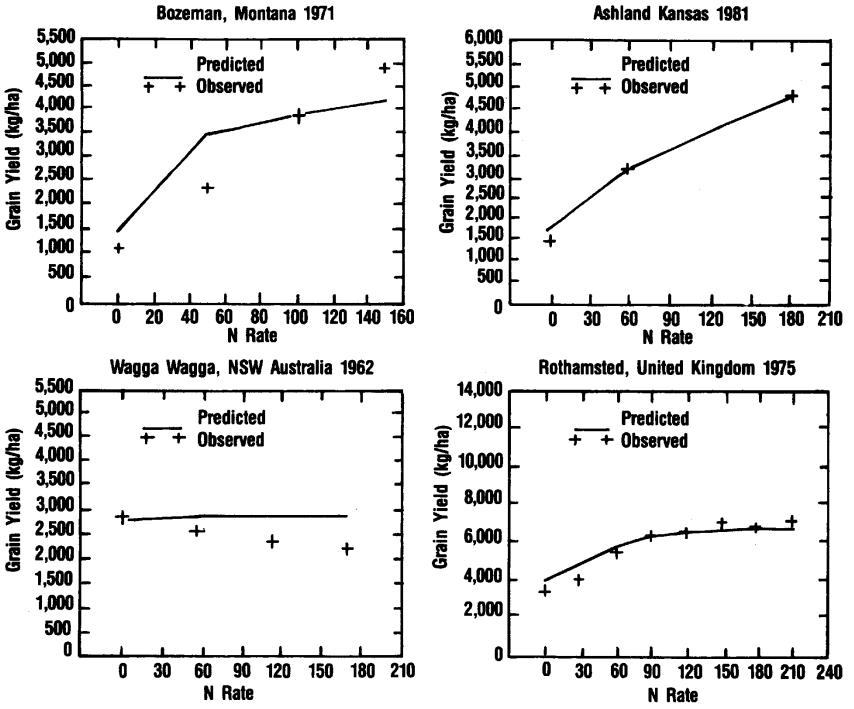


Fig. 13–7. Comparison of predicted (lines) and observed (pluses) grain yield response to applied N in individual data sets using the CERES-Wheat model. (a) Bozeman, MT (observed data from Christensen & Killorn, 1981); (b) Ashland, KS (observed data from Wagger, 1983); (c) Wagga Wagga, NSW, Australia (observed data from Storrier, 1965); and (d) Rothamsted, U.K. (observed data from Pearman et al., 1978).

operate in a one-dimensional manner, placement patterns such as side banding and point placement can only be simulated by assuming uniform incorporation into a layer. Under certain conditions this may lead to erroneous predictions. These various problems will be the subject of future efforts to further enhance the model's versatility. From the validation studies cited above, it appears that the model is able to explain most of the observed variation in yield and other key parameters where these conditions of loss and placement do not apply.

The model, particularly when coupled with long-term weather data or generated climatic data, is a valuable tool for providing insights into the behavior of many aspects of a cropping system. Running the model with long-term weather data enables a quantification of the temporal variability in yield and response to fertilizer. The frequency and nature of N losses from the system leading to poor fertilizer efficiency can be identified and their rela-

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tive significance evaluated. Using the model, experiments can be performed to predict the effects of varying fertilizer rates, timing, placement depths, fertilizer sources, planting times, etc. In this way, a fertilizer strategy to maximize efficiency under the uncertainties of climatic variability can be readily obtained. Studies such as this have been reported by Godwin and Vlek (1985).

V. ACKNOWLEDGMENTS

The authors wish to gratefully acknowledge the researchers who generously provided unpublished data for model development and testing.

VI. APPENDIX 1

A listing of the four subroutines used in the CERES models for simulating N transformations, uptake, movement, and deficiency factors.

```
C*****************DRAINAGE AND LEACHING ROUTINE*********************
C
      SUBROUTINE NFLUX(iflag,skgn,sppm,fac)
C
$Include: 'Whea3.Blk'
$Include: 'Nmove.Blk'
C
      Dimension skgn(15),sppm(15),fac(15)
      if(iflag.eq.1) go to 300
      KPFLG=0
90    OUTN=0.0
      DO 200 L=1,NLAYR
           skgn(L)=skgn(L)+OUTN
           sppm(L)=skgn(L)*FAC(L)
           xmin=0.25/fac(1)
 100       OUTN=(skgn(L)-xmin)*FLUX(L)/(SW(L)*DLAYR(L)+FLUX(L))
           IF(KPFLG.EQ.1)OUTN=OUTN*0.5
           skgn(L)=skgn(L)-OUTN
           sppm(L)=skgn(L)*FAC(L)
           If(L.EQ.Nlayr) Tlch=Tlch+outn
 200  CONTINUE
      RETURN
 300  continue
      OUTN=0.0
      KPFLG=0
      DO 500 J=1,MU
           K=MU+1-J
           skgn(K)=skgn(K)+OUTN
           IF(K.EQ.1) GO TO 500
           IF(FLOW(K-1).GE.0.0)GO TO 400
           OUTN=0.0
           KPFLG=1
           GO TO 500
 400       xmin=0.25/fac(k)
           OUTN=(skgn(K)-xmin)*(FLOW(K-1)/(SW(K)*DLAYR(K)+FLOW(K)))*0.5
           skgn(K)=skgn(K)-OUTN
 500  CONTINUE
      IF(KPFLG.EQ.0)RETURN
      OUTN=0.0
      DO 700 L=1,NLAYR
```

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```
          FLUX(L)=0.0
          IF(FLOW(L).LT.0.0)FLUX(L)=FLOW(L)*(-1.0)
700     CONTINUE
        GO TO 90
        END
C
C*********************NITROGEN DEFICIENCY FACTOR ROUTINE*************
C
        SUBROUTINE NFACTO
C
$Include: 'Whea1.Blk'
$Include: 'Whea2.Blk'
$Include: 'Ntrc1.Blk'
C
        REAL NFAC
C***** CONVERT XSTAGE TO ZADOKS GROWTH STAGE
        IF(XSTAGE.LE.2.0)ZSTAGE=XSTAGE
        IF(XSTAGE.GT.2..AND.XSTAGE.LE.3.0)ZSTAGE=2.0+2.0*(XSTAGE-2.0)
        IF(XSTAGE.GT.3..AND.XSTAGE.LE.4.)ZSTAGE=4.0+1.7*(XSTAGE-3.0)
        IF(XSTAGE.GT.4..AND.XSTAGE.LE.4.4)ZSTAGE=5.7+0.8*(XSTAGE-4.0)
        IF(XSTAGE.GT.4.4.AND.XSTAGE.LE.6.)ZSTAGE=6.02+1.8625*(XSTAGE-4.4)
        YSTAGE=XSTAGE
        zs2=zstage*zstage
        zs3=zs2*zstage
        zs4=zs3*zstage
        IF(P1V.ge.0.03)THEN
          TCNP=-5.0112400-6.350677*ZSTAGE+14.95784*SQRT(ZSTAGE)
       1 +0.2238197*ZS2
        ELSE
          TCNP=7.4531813-1.7907829*ZSTAGE+0.6092849*SQRT(ZSTAGE)
       1 +0.0933967*ZS2
        ENDIF
        IF(ZSTAGE.GT.6.0)TCNP=TCNP-(ZSTAGE-6.0)*0.140
        TCNP=TCNP/100.0
        TMNC=(2.97-0.455*XSTAGE)/100.0
C        RCNP=(-0.25*XSTAGE+2.5)/100.0
        rcnp=(2.10-0.14*sqrt(zstage))*0.01
        IF (ISTAGE.LT.5) GO TO 5
        TANC=VANC
        VMNC=TMNC
5       NFAC=1.0-(TCNP-TANC)/(TCNP-TMNC)
        if(nfac.lt.0.02)nfac=0.02
        IF (XSTAGE.LE.1.10) NFAC=1.0
        ndef4=nfac*nfac
        if(ndef4.gt.1.5)ndef4=1.5
        IF (TANC.GE.TCNP) NFAC=1.0
        NDEF3=NFAC*NFAC
        NDEF1=0.10+2.*NFAC
        NDEF2=NFAC
        IF(NDEF2.GT.1.0)NDEF2=1.0
        IF(NDEF1.GT.1.0)NDEF1=1.0
        IF(NDEF3.GT.1.0)NDEF3=1.0
        CNSD1=CNSD1+1.0-Ndef1
        CNSD2=CNSD2+1.0-NDEF2
        RETURN
C
10      FORMAT (' EXTREME NITROGEN DEFICIENCY INCURRED ON DAY',I4)
        END
C
C
C
C
C
C
        SUBROUTINE SOLT
C
```

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```
$Include: 'Whea1.blk'
$Include: 'Whea4.Blk'
$Include: 'Whea3.Blk'
$Include: 'Ntrc2.Blk'
      XI=DOY
      ALX=(XI-HDAY)*0.0174
      ATOT=ATOT-TMA(5)
      K=5
  100 TMA(K)=TMA(K-1)
      K=K-1
      IF (K.GT.1) GO TO 100
      TMA(1)=(1.-ALBEDO)*(Tempm+(TEMPMX-Tempm)*SQRT(SOLRAD*.03))
     1 +ALBEDO*TMA(1)
      ATOT=ATOT+TMA(1)
      AW=PESW
      IF(AW.LE.0.0)AW=0.01
      WC=AW/(WW*depmax*10.)
      F=EXP(B*((1.-WC)/(1.+WC))**2)
      DD=F*DP
      TA=TAV+AMP*COS(ALX)/2.
      DT=ATOT/5.-TA
      DO 200 L=1,NLAYR
           ZD=-Z(L)/DD
           ST(L)=TAV+(AMP/2.*COS(ALX+ZD)+DT)*EXP(ZD)
  200 CONTINUE
      RETURN
      END
C

      SUBROUTINE NUPTAK
C
$Include: 'Whea1.blk'
$Include: 'Whea2.Blk'
$Include: 'Whea3.Blk'
$Include: 'Whea4.Blk'
$Include: 'Ntrc1.Blk'
$Include: 'Ntrc2.Blk'
$Include: 'Nmove.Blk'
C
      REAL NUF,ndem
      DIMENSION RNO3U(15), RNH4U(15)
      TRNLOS=0.0
      ranc=rootn/rtwt
      tanc=topsn/topwt
      trlv=0.
      Ano3=0.0
      Anh4=0.0
      DO 100 L=1,NLAYR
           rno3u(l)=0.
           rnh4u(l)=0.
           trlv=trlv+rlv(l)
           NO3(L)=SNO3(L)*FAC(L)
           NH4(L)=SNH4(L)*FAC(L)
           Ano3=ano3+Sno3(l)
           Anh4=anh4+Snh4(L)
           TOTN=NO3(L)+NH4(L)
  100 CONTINUE
      IF (PDWI.EQ.0.) PDWI=1.
      DNG=PDWI*TCNP
      IF (XSTAGE.LE.1.2) DNG=0.0
      TNDEM=TOPWT *(TCNP-TANC)+DNG
      RNDEM=RTWT*(RCNP-RANC)+PGRORT*RCNP
      NDEM=TNDEM+RNDEM
      ANDEM=NDEM*PLANTS*10.0
      DROOTN=0.0
      DSTOVN=0.0
```

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```
      TRNU=0.0
      TNUP=0.0
      nuf=0.
      DO 200 L=1,NLAYR
           IF (RLV(L).EQ.0.0) GO TO 300
           L1=L
           FNH4=1.0-EXP(-0.025*(NH4(L)-0.5))
           FNO3=1.0-EXP(-0.0275*NO3(L))
           IF (FNO3.LT.0.03) FNO3=0.0
           IF (FNO3.GT.1.0) FNO3=1.0
           IF (FNH4.LT.0.03) FNH4=0.0
           IF (FNH4.GT.1.0) FNH4=1.0

                SMDFR=(SW(L)-LL(L))/ESW(L)
                if(smdfr.lt.0.0)smdfr=0.
                if(smdfr.gt.1.0)smdfr=(sat(l)-sw(l))/(sat(l)-dul(l))
                RFAC=RLV(L)*SMDFR*SMDFR*DLAYR(L)*100
      c          if(xstage.gt.5.0)rfac=rfac*(6.0-xstage)
                RNO3U(L)=RFAC*FNO3*0.009
                RNH4U(L)=RFAC*FNH4*0.009
                TRNU=TRNU+RNO3U(L)+RNH4U(L)
        200 CONTINUE
        300 If(andem.le.0.)then
               Trnu=0.
               Nuf=0.
               Go to 400
           Endif
           IF (ANDEM.GT.TRNU) ANDEM=TRNU
           IF (TRNU.EQ.0.0) GO TO 600
           NUF=ANDEM/TRNU
            Trnu=0.0
        400 Continue
           DO 500 L=1,L1
                UNO3=RNO3U(L)*NUF
                UNH4=RNH4U(L)*NUF
                Xmin=0.25/FAC(L)
                if(uno3.gt.sno3(l)-xmin)uno3=sno3(l)-xmin
                SNO3(L)=SNO3(L)-UNO3
                Xmin=0.5/FAC(L)
                if(unh4.gt.(snh4(l)-xmin))unh4=snh4(l)-xmin
                SNH4(L)=SNH4(L)-UNH4
                NO3(L)=SNO3(L)*FAC(L)
                NH4(L)=SNH4(L)*FAC(L)
                Rnloss=0.0
                if(tanc.gt.tcnp)rnloss=ranc*rtwt*0.05*plants*rlv(l)/trlv
                TRNLOS=TRNLOS+RNLOSS
                fon(l)=fon(l)+rnloss
                trnu=trnu+uno3+unh4
        500 CONTINUE
           TRNU=TRNU/(PLANTS*10.0)
           if(ndem.gt.trnu)Then
               xndem=trnu
               Factor=xndem/ndem
               ndem=xndem
               tndem=tndem*factor
               rndem=rndem*factor
           Endif
           DTOPSN=TNDEM/NDEM*TRNU-ptf*trnlos/(plants*10)
           DROOTN=RNDEM/NDEM*TRNU-(1.0-ptf)*trnlos/(plants*10)
           TOPSN=TOPSN+DTOPSN
           TANC=TOPSN/TOPWT
           ROOTN=ROOTN+DROOTN
           RANC=ROOTN/(RTWT-0.01*RTWT)
           tno3=0.0
           tnh4=0.0
           do 599 l=1,nlayr
```

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```
        tno3=tno3+sno3(l)
        tnh4=tnh4+snh4(l)
599   Continue
  600 RETURN
      END
C
C
C
C
C******************N Transformations Subroutine************************
C
      SUBROUTINE NTRANS
C
$Include: 'Whea1.blk'
$Include: 'Whea3.Blk'
$Include: 'Whea4.Blk'
$Include: 'Ntrc1.Blk'
$Include: 'Ntrc2.Blk'
$Include: 'Nmove.Blk'
      Real Nnom,Mf
c
      Dimension rdecr(3),PROF(15)
      data rdecr/0.2,0.05,0.0095/
      tifon=0.0
      thumn=0.0
      IF (nFERT .eq. 0) GO TO 900
      DO 10 L=1,NLAYR
         PROF(L)=0.
10    CONTINUE
      DEPTH=0.0
      KMAX=1
      DO 100 J=1,NFERT
         IF (DOY .eq. Fday(J)) THEN
           M=IFTYPE(J)
           IF(M.EQ.17) M=12
           KMAX=IDLAYR(NLAYR,DLAYR,DFERT(J))
           CD=0.0
           IF(KMAX.EQ.1) THEN
              PROF(1)=1.0
C          *** Surface layer incorporation ***
           ELSE
              CD=DLAYR(1)
              PROF(1)=DLAYR(1)/DFERT(J)
              DO 20 L=2,KMAX
                 CD=CD+DLAYR(L)
                 IF (DFERT(J).LE.CD) THEN
                    PROF(L)=(DFERT(J)-(CD-DLAYR(L)))/DFERT(J)
                 ELSE
                    PROF(L)=DLAYR(L)/DFERT(J)
                 ENDIF
20            CONTINUE
           ENDIF
           GO TO (500,400,500,400,300,400,400,600,400,700,400,600), M
C          FERTILIZER TYPES AS GIVEN IN APPENDIX 4, TECHNICAL REPORT 1,
C            IBSNAT (1986).
C            1    =AMMONIUM NITRATE
C            2    =AMMONIUM SULPHATE
C            3    =AMMONIUM NITRATE-SULPHATE
C            4    =ANHYDROUS AMMONIA
C            5    =UREA (HYDROLYSIS TO BE ADDED LATER)
C            6    =DIAMMONIUM PHOSPHATE
C            7    =MONOAMMONIUM PHOSPHATE
C            8    =CALCIUM NITRATE
C            9    =AQUA AMMONIA
C           10    =UREA AMMONIUM NITRATE
```

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```
C            11   =CALCIUM AMMONIUM NITRATE
C            17   =POTASSIUM NITRATE
C
  300       DO 350 K=1,KMAX
              UREA(K)=UREA(K) + AFERT(J)*PROF(K)
  350       CONTINUE
            IUON=.TRUE.
            IUOF=DOY + 21
            IF(DOY.GT.344) IUOF=21-(365-DOY)
            GO TO 900
  400       DO 450 K=1,KMAX
              SNH4(K)=SNH4(K)+AFERT(J)*PROF(K)
  450       CONTINUE
            GO TO 900
  500       DO 550 K=1,KMAX
              SNH4(K)=SNH4(K)+0.5*AFERT(J)*PROF(K)
              SNO3(K)=SNO3(K)+0.5*AFERT(J)*PROF(K)
  550       CONTINUE
            GO TO 900
  600       DO 650 K=1,KMAX
              SNO3(K)=SNO3(K)+AFERT(J)*PROF(K)
  650       CONTINUE
            GO TO 900
  700       DO 750 K=1,KMAX
              SNO3(K)=SNO3(K)+AFERT(J)*0.25*PROF(K)
              SNH4(K)=SNH4(K)+AFERT(J)*0.25*PROF(K)
              UREA(K)=UREA(
                                K)+AFERT(J)*0.50*PROF(K)
  750       CONTINUE
            IUON=.TRUE.
            IUOF=DOY + 21
            IF(DOY.GT.344) IUOF=21-(365-DOY)
            GO TO 900
          ENDIF
  100   CONTINUE
  900   CONTINUE
        IF(DOY.EQ.IUOF) THEN
C ***** Assume all urea was hydrolyzed.
            DO 1000 L=1,NLAYR
                 SNH4(L)=SNH4(L) + UREA(L)
                 UREA(L)=0.0
                 UPPM(L)=0.0
 1000       CONTINUE
            IUON=.FALSE.
        ENDIF
        DO 1500 l=1,NLAYR
            ad=ll(l)
            if(l.eq.1)ad=ll(l)*swef
            mf=(sw(l)-ad)/(dul(l)-ad)
            wfd=mf
            if(sw(l) .gt. dul(l))then
                 xl=(sw(l)-dul(l))/(sat(l)-dul(l))
                 mf=1.0-0.5*xl
                 wfd=1.0-xl
            endif
            if(wfd .lt. 0.0)wfd=0.0
            NO3(l)=SNO3(l)*FAC(l)
            NH4(l)=SNH4(l)*FAC(l)
            UPPM(L)=UREA(L)*FAC(L)
            IF(IUON) THEN
C ***** DO SOME UREA HYDROLYSIS.
                 SWF=MF+0.20
                 IF(SWF.GT.1.0) SWF=1.0
                 if(swf.lt.0.) swf=0.
                 TF=(ST(L)/40.0)+0.20
```

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```
                IF(TF.LT.0) TF=0.0
                AK=-1.12+1.31*OC(L)+0.203*PH(L)-0.155*OC(L)*PH(L)
                IF(AK.LT.0.25) AK=0.25
                UHYDR=AK*AMIN1(SWF,TF)*UREA(L)
                IF(UHYDR.GT.UREA(L)) UHYDR=UREA(L)
                UREA(L)=UREA(L)-UHYDR
                SNH4(L)=SNH4(L)+UHYDR
              ENDIF
              tf=(st(l)-5.0)/30.0
              if(st(l) .lt. 5.0)tf=0.0
              TOTN=SNO3(l)+SNH4(l)-0.5/FAC(l)
              IF (TOTN .lt. 0.0) TOTN=0.0
              CNR=(0.4*FOM(l))/(FON(l)+TOTN)
              CNRF=EXP(-0.693*(CNR-25)/25.0)
              IF (CNRF .gt. 1.0) CNRF=1.0
              fom(l)=0.0
              grcom=0.0
              grnom=0.0
              if(mf.lt.0.)mf=0.
              do 1200 jp=1,3
                   fom(l)=fom(l)+fpool(l,jp)
                   if(fom(l).lt.5.0)then
                        g1=0.0
                        go to 1100
                   endif
                   g1=tf*mf*cnrf*rdecr(jp)
                   x=fpool(l,jp)/fom(l)
                   fpool(l,jp)=fpool(l,jp)-fpool(l,jp)*g1
1100               grcom=grcom+g1*x*fom(l)
                   grnom=grnom+g1*x*fon(l)
1200          continue
              RHMIN=NHUM(l)*DMINR*TF*MF*dmod
              HUM(l)=HUM(l)-RHMIN*10.0+0.2*GRNOM/0.04
              NHUM(l)=NHUM(l)-RHMIN+0.2*GRNOM
              RNAC=AMIN1(TOTN,grcom*(0.02-FON(l)/FOM(l)))
              IF(RNAC .lt. 0.)RNAC=0.
              FOM(l)=FOM(l)-grcom
              FON(l)=FON(l)+RNAC-GRNOM
              NNOM=0.8*GRNOM+RHMIN-RNAC
              tifon=tifon+fon(l)
              thumn=thumn+nhum(l)
              XMIN=0.5/FAC(L)
c     net mineralization added here
c  immobilization if nnom is less than 0.
c   first got N from ammonium- when too low get remainder from nitrate
              if(nnom.gt.0)then
                   snh4(l)=snh4(l)+nnom
              else
                   if(abs(nnom).gt.(snh4(l)-xmin))then
                      nnom=nnom+snh4(l)-xmin
                      Snh4(l)=xmin
                   else
                       Snh4(l)=snh4(l)+nnom
                       nnom=0.0
                   endif
                   Sno3(l)=sno3(l)+nnom
               Endif
              tmin=tmin+nnom
c             c*************nitrification section
 1300         sanc=1.0-exp(-0.01363*snh4(l))
              ELNC=AMIN1(TF,WFD,SANC)
              RP2=CNI(l)*EXP(2.302*ELNC)
              IF (RP2 .lt. 0.05) RP2=0.05
              IF (RP2 .gt. 1.0) RP2=1.0
              CNI(l)=RP2
```

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```
            A=AMIN1(RP2,WFD,TF,phn(l))
            NH4(l)=SNH4(l)*FAC(l)
            bb=(a*40.0*NH4(l)/(NH4(l)+90.0))
            if(bb.gt.0.80)bb=0.80
            bb=bb*snh4(l)
            b2=(nh4(l)-0.5)/fac(l)
             rntrf=amin1(bb,b2)
           SNH4(l)=SNH4(l)-RNTRF
           SNO3(l)=SNO3(l)+RNTRF
           SARNC=1.0-EXP(-0.1363*SNH4(l))
           XW=AMAX1(WFD,WFY(l))
           XT=AMAX1(TF,TFY(l))
           CNI(l)=CNI(l)*AMIN1(XW,XT,SARNC)
           IF (CNI(l) .le. 0.05) CNI(l)=0.05
           WFY(l)=WFD
           TFY(l)=TF


c*********denitrification section
            dnrate=0.0
            NO3(l)=SNO3(l)*FAC(l)
            IF(NO3(L).lt.1.0.OR.SW(l).le.dul(l).or.st(l).lt.5) GO TO 1400
            FW=0.0
            SOILC=0.58*HUM(L)
            CW=FAC(L)*(SOILC*0.0031+0.4*fpool(l,1))+24.5
            FW=1.0-(SAT(L)-SW(L))/(SAT(L)-DUL(L))
            FT=0.1*EXP(0.046*ST(L))
            DNRATE=6.0*1.E-05*CW*NO3(L)*BD(L)*FW*FT*DLAYR(L)
c c           If(Dnrate.lt.0.) Dnrate=0.0
            xmin=0.25/fac(l)
            IF(DNRATE.GT.(SNO3(L)-xMIN)) DNRATE=SNO3(L)-xMIN
            SNO3(L)=SNO3(L)-DNRATE
            Tnox=Tnox+dnrate
 1400       Continue
            NO3(L)=SNO3(L)*FAC(L)
            NH4(L)=SNH4(L)*FAC(L)
 1500 continue
      RETURN
      END
C*****
      Function Idlayr(Nlayr,Dlayr,Fdepth)
      Dimension Dlayr(Nlayr)
c****** routine used to identify layer where fertilizer is placed
       Depth=0
       Idlayr=1
       Do 10 l=1,nlayr
       Depth=Depth+Dlayr(l)
       If(Fdepth.gt.Depth)go to 10
       Idlayr=L
       Return
10     Continue
       Idlayr=Nlayr
       Return
       End
```

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VII. APPENDIX 2

Glossary of the variables used in the N subroutines of the CERES models.

A	Zero to unity factor for relative nitrification rate (unitless)
AFERT(J)	Amount of N added as fertilizer on JFDAY(J) (kg N/ha)
AK	Nondimensional constant defining proportion of urea which hydrolyzes in 1 d
ANDEM	Crop N demand (kg N/ha)
BB	Maximum rate of nitrification (kg N/ha per day)
BD	Bulk density of soil (g/cm**3)
CNI(L)	Capacity for nitrification index in layer L. This is a zero to unity number indicating the relative capability for nitrification to proceed
CNR	C/N ratio calculated as (kg C in FOM)/(kg N in FOM + kg mineral N)
CNRF	Zero to unity C/N ratio factor for decomposition rate
CNSD1	Accumulates N deficit factor (NFAC) in each state and is printed at the end of each stage as a daily average
CNSD2	Accumulates N deficit factor (NDEF2) in each stage and is printed at the end of each stage as a daily average
CW	Water soluble carbon content of soil (ppm)
DEF	Interim variable used to ensure soil N pools remain positive
DEPTH	Depth to the bottom of a layer from the surface (cm)
DEFERT(J)	Depth of incorporation of fertilizer application on Julian date (JFD)
DLAYR(L)	Depth increment of soil layer L − cm
DMINR	Humic fraction decay rate (1/days)
DMOD	Zero to unity dimensionless factor used to decrease to rate of mineralization in soils with chemically protected organic matter
DNG	N demand of potential new growth of tops (g N/plant)
DNRATE	Denitrification rate (kg N/ha per day)
DOY	Day of the year
DROOTN	Daily change in plant root N content (g N/plant)
DTOPSN	Daily change in plant tops N content (g N/plant)
DUL(L)	Drained upper limit soil water for soil layer L − volume fraction
ELNC	Environmental limit on nitrification capacity (zero to unity unitless factor)
ESW(L)	Extractable soil water content for soil layer L (the difference between DUL and LL − volume fraction)
FAC(L)	Conversion factor for ppm N to kg N/ha for layer L
FDAY(J)	Day of the year when fertilizer application J was made
FLOW(L)	Volume of water moving from layer L due to unsaturated flow (CM) positive indicates upward movement and negative value indicates downward movement
FLUX(L)	Water moving downward from layer L with drainage (cm)
FNH4	Unitless soil ammonium supply index
FNO3	Unitless soil nitrate supply index
FOM(L)	Fresh organic matter (residue) in layer L (kg/ha)
FON(L)	N in fresh organic matter in layer L (kg N/ha)
FPOOL(L,J)	Fresh organic matter in layer L (kg OM/ha). If J = 1 pool is comprised of carbohydrates, if J = 2 pool is comprised of cellulose, and if J = 3 pool is comprised of lignin
FT	Temperature factor affecting denitrification rate
FW	Unitless soil moisture factor affecting denitrification rate

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G1	Genetic specific constant related to rate of vegetative expansion growth during Stage 1
GRCOM	Gross release of carbon from organic matter decomposition (kg C/ha)
GRNOM	Gross release of N from organic matter decomposition (kg N/ha per day)
GRORT	Daily root growth (g/day)
HUM(L)	Stable humic fraction material in layer L (kg/ha)
IFLAG	Switch variable used to direct control to either the leaching component or the upward flux component of subrouting NFLUX
IFTYPE	Code number for fertilizer type
ISTAGE	Phenological stage = 1 Emergence to terminal spikelet = 2 Terminal spikelet to end of vegetative growth = 3 End of vegetative growth to end of pre-anthesis ear growth = 4 Pre-anthesis ear growth to beginning of grain fill (anthesis occurs during this phase) = 5 Beginning of grain fill to physiological maturity = 6 Physiological maturity to fallow (harvest) = 7 Fallow to sowing = 8 Sowing to germination = 9 Germination to emergence
IUOF	Switch variable to indicate urea hydrolysis is not active
IUON	Switch variable to indicate urea hydrolysis is active
L1	The number of soil layers to the bottom of the root zone
LL(L)	Lower limit soil water content for soil layer L — volume fraction
MF	Zero to unity moisture factor for residue decomposition rate
MU	Loop variable to indicate layer below the current layer
NDEF1	Zero to unity N deficiency factor for photosynthetic rate
NDEF2	Zero to unity N deficiency factor for expansion growth
NDEF3	Zero to unity N deficiency factor for tiller number
NDEF4	Zero to unity N deficiency factor for grain N determination
NDEM	Plant N demand (g/plant)
NFAC	Zero to unity factor based on actual and critical N concentrations
NFERT	Number of fertilizer applications made
NH4(L)	Soil ammonium (ppm) in layer L
NHUM(I)	N associated with the stable humic fraction in layer I (kg N/ha)
NLAYR	Number of layers in soil
NNOM	Net N released from all organic sources in a layer (kg N/ha)
NO3(L)	Soil nitrate (ppm) in layer L
NUF	Plant N supply/demand ratio used to modify uptake
OC(L)	Organic carbon in layer L (%)
OUTN	Nitrate N leaching from a layer (kg N/ha)
PIV	Genetic specific coefficient that determines sensitivity to vernalization
PDWI	Potential increment of new shoot growth (g/plant)
PGRORT	Potential increment of new root growth (g/plant)
PH(L)	Soil pH in layer L
PHN(L)	Zero to unity factor describing the effect of soil pH or nitrification rate on layer L
PLANTS	Number of plants per square meter
PTF	Fraction of photosynthesis partitioned to aboveground plant parts
RANC	Root actual N concentration (g N/g root dry weight)

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RCNP	Root critical N concentration (g N/g root dry weight)
RDECR(J)	The maximum rate constant for decay of residue components (1/days)
RFAC	Interim variable describing the effects of root length density on potential N uptake from a layer
RHMIN	N mineralized from humus in a layer (kg N/ha)
RLV(L)	Root length per unit soil volume for layer L (cm/cm^3)
RNAC	Immobilization rate of N associated with the decay of residues (kg N/ha per day)
RNDEM	Plant root demand for N (g/plant)
RNH4U(L)	Potential ammonium uptake from layer L (kg N/ha)
RNLOSS	Loss of N from the plant via root exudation in one layer (g N/plant)
RNO3U(L)	Potential nitrate uptake from layer L (kg N/ha)
RNTRF	Amount of ammonium nitrified in a layer (kg N/ha per day)
ROOTN	Plant root N content (g N/plant)
RP2	Temporary variable used in nitrification calculations
RTWT	Root weight (g/m^2)
SANC	Supply of ammonium effect on nitrification capacity
SARNC	Supply of ammonium effect on the reduction of nitrification capacity (zero to unity, unitless)
SAT(L)	Field saturated soil water content in layer L (cm/cm)
SKGN(L)	Dummy variable for either urea or nitrate present in layer L (kg N/ha)
SMDER	Soil moisture deficit factor affecting N uptake
SMIN	Interim variable to prevent soil N pools from becoming <1 ppm
SNH4(L)	Soil ammonium in layer L (kg N/ha)
SNO3(L)	Soil nitrate in layer L (kg N/ha)
SOILC	Soil carbon content (kg C/ha)
SPPM(L)	Dummy variable for either urea or nitrate concentration present in layer L (ppm N)
ST(L)	Soil temperature in layer L (°C)
SW(L)	Actual soil water content in layer L (cm/cm)
TANC	Tops actual N concentration (g N/g dry wt.)
TCNP	Tops critical N concentration (g N/g dry wt.)
TF	Temperature factor for nitrification on mineralization
TEY(L)	Yesterday's temperature factor for nitrification in layer L
TMNC	Plants tops minimum N concentration (g N/g dry wt.)
TNDEM	Plant tops demand for N (g N/plant)
TNUP	Total N uptake from the profile on 1 d (kg N/ha)
TOPSN	N contained in plant tops excluding grain (g N/plant)
TOPWT	Weight of plant tops excluding grain (g)
TOTN	Total mineral N in a layer (kg N/ha)
TRLV	Total root length density variable
TRNLOS	Total plant N lost by root exudation (g N/plant)
TRNU	Total potential root nitrogen uptake from the soil (kg N/ha)
UHYDR	Urea hydrolysis rate (kg urea N/ha per day)
UNH4	Plant uptake of ammonium from a layer (kg N/ha)
UNO3	Plant uptake of nitrate from a layer (kg N/ha)
UP1	Interim variable used to prevent soil N pools from becoming <1 ppm
UPPM(L)	Concentration of urea N in layer L (ppm)
UREA(L)	Urea N in layer L (kg N/ha)
VANC	Plant vegetative actual N concentration (g N/plant)
VMNC	Plant vegetative minimum N concentration (g N/g dry wt.)

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WFD	Today's water factor for nitrification
WEY(L)	Yesterday's water factor for nitrification in layer L
XL	Temporary variables used to determine soil
XSTAGE	Non-integer growth stage indicator ranging from zero to six
XT	Temperature effect on nitrification capacity
XW	Moisture effect on nitrification capacity
ZS2	Interim variables used in critical concentration calculations
ZS3	Interim variables used in critical concentration calculations
ZSTAGE	Zadoks' growth stage

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